

Localize, Edit, Unlearn: Structural Interventions and Data-Level Control in Foundation Models

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Ph.D. Thesis Defense

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Research Theme

Knowledge Localization
& Model Editing

Data Interpretability &
Machine Unlearning

Generative AI: 2022 – Now

Text-to-Image Models

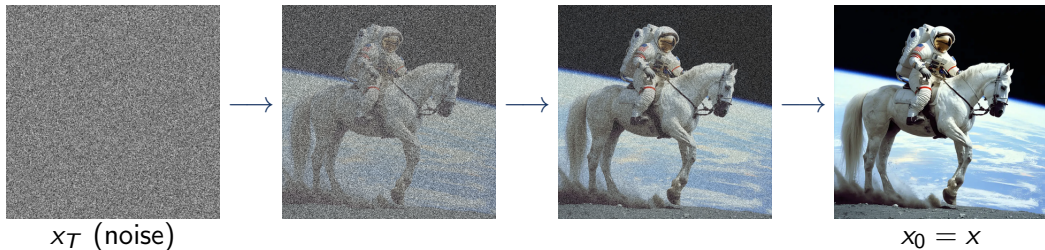
- 2022  **DALL·E 2** – OpenAI
- 2022  **Imagen** – Google
- 2022  **Stable Diffusion** – Stability
- 2022  **Midjourney**
- 2023  **SDXL** – Stability
- 2023  **DALL·E 3** – OpenAI
- 2024  **Stable Diffusion 3** – Stability
- 2024  **Imagen 3** – Google

Large Language Models

- 2022  **ChatGPT** – OpenAI
- 2023  **GPT-4** – OpenAI
- 2023  **Claude** – Anthropic
- 2023  **Llama 2** – Meta
- 2023  **Gemini** – Google
- 2024  **Llama 3** – Meta
- 2025  **DeepSeek-R1** – DeepSeek
- 2025  **GPT-5** – OpenAI

Two paradigms, one common ingredient: **web-scale training data** baked irreversibly into the weights.

Text-to-Image: Diffusion



"an astronaut riding a horse in space" = c



$$\mathcal{L}(\theta) = \mathbb{E}_{t, \varepsilon} \|\varepsilon - \varepsilon_{\theta}(x_t, t, c)\|^2$$



U-Net $\rightarrow$ Transformer

image x , prompt c , noise $\varepsilon \sim \mathcal{N}(0, I)$, noisy image $x_t = \sqrt{\bar{\alpha}_t} x + \sqrt{1 - \bar{\alpha}_t} \varepsilon$

Large Language Models: Next-Token Prediction

The capital of France is __

Paris 0.92

Lyon 0.03

the 0.02

a 0.01


$$\mathcal{L}(\theta) = - \sum_{t=1}^T \log p_{\theta}(x_t \mid x_1, \dots, x_{t-1})$$
 Transformer

tokens $x_1, x_2, \dots, x_T$ of a training sequence

Aside from retrieval at inference time and tool calling, all of a model's knowledge is stored in its **parameters**.

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Understanding Model

Where is the knowledge?
Can we localize it?

Understanding Data

How does a training sample
affect the model?
Can we measure that?

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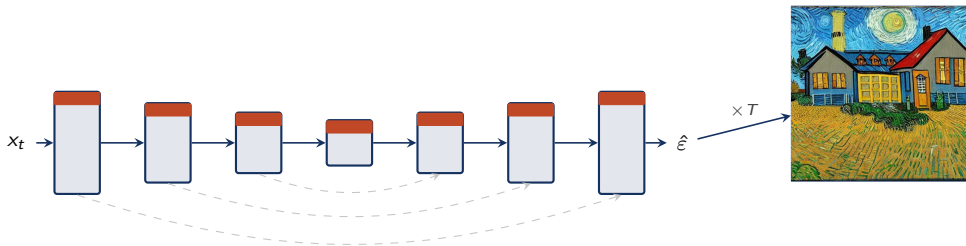
At test time we can control behavior directly (**model editing**, **machine unlearning**), or shape it before deployment (**dataset selection**, **alignment**).

Research Theme

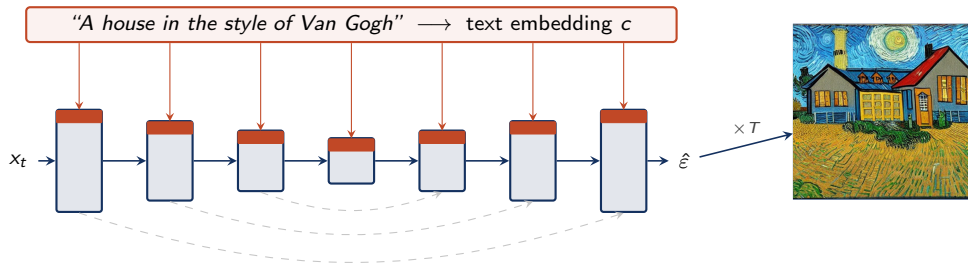
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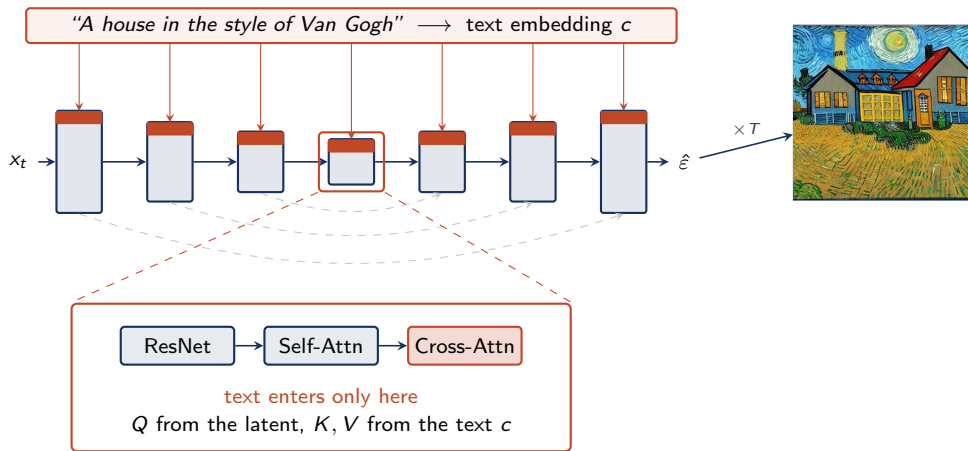
Inside a U-Net Diffusion Model: Stable Diffusion 2 (2022)



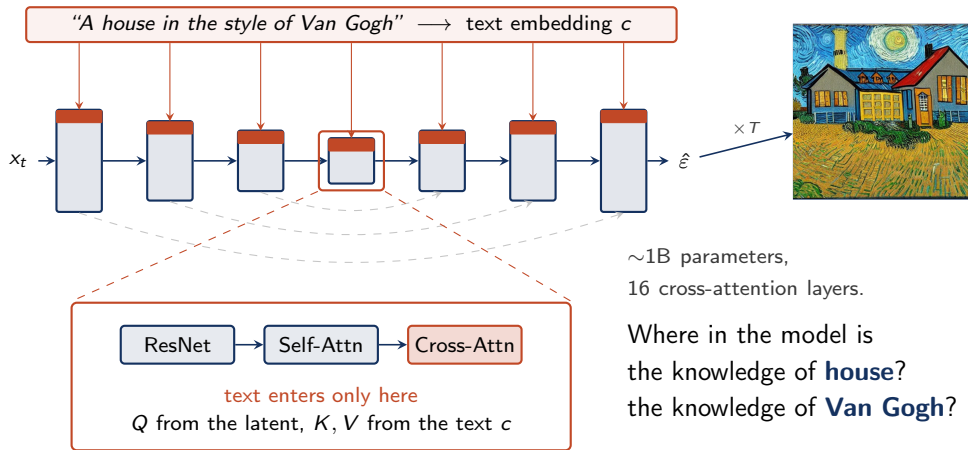
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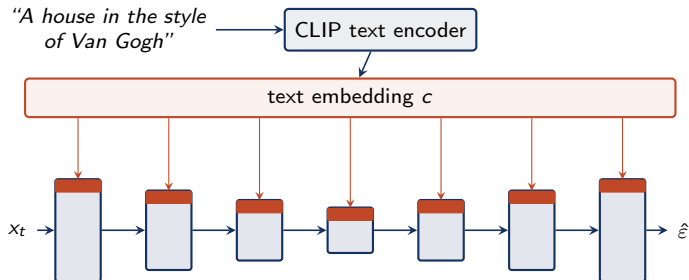
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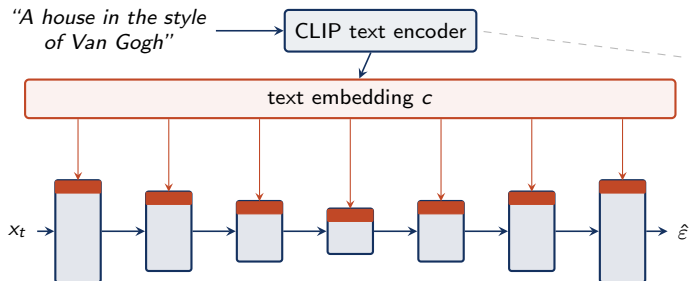
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Are All Layers Equal?

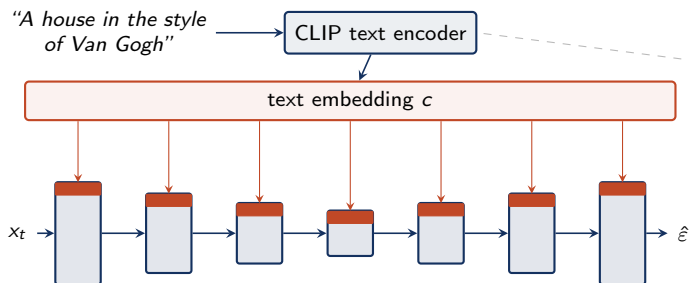


Are All Layers Equal?



Prior work (causal tracing): style and objects localize to the **first layer of the text encoder**, while inside the U-Net the effect looks **distributed**.

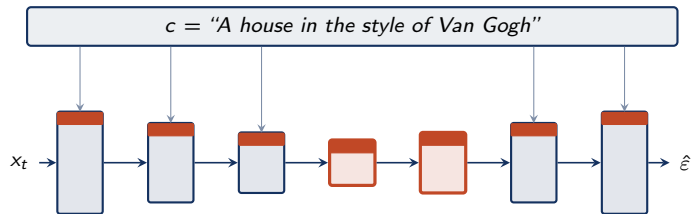
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All **16 cross-attention layers** receive the *same* c .
Are all of them equally responsible?

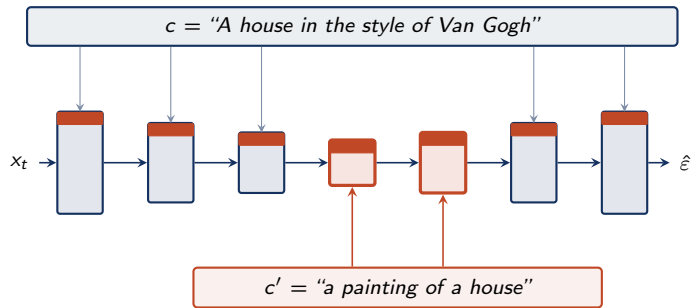
LocoGen: Give a Subset a Different Prompt



all layers get c
style present

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LocoGen: Give a Subset a Different Prompt



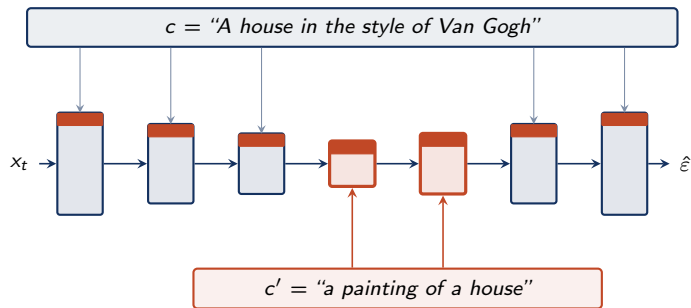
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LocoGen: Give a Subset a Different Prompt



all layers get c
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subset gets c'
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The style disappears when only **a few adjacent layers** see c' .

Is that where "Van Gogh" is stored?

Basu, Rezaei, et al. *On Mechanistic Knowledge Localization in Text-to-Image Generative Models*. ICML 2024.

LocoGen: Searching for the Controlling Layers

Which m of the M cross-attention layers should get c' ?

Brute force tries all $\binom{M}{m}$ subsets (SD-XL: 12M).

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LocoGen

```
for  $j = 1, \dots, M - m + 1$ :  
     $C' \leftarrow \{C_j, \dots, C_{j+m-1}\}$  a window of  $m$  adjacent layers  
    generate  $N$  images: layers in  $C'$  receive  $c'_i$ , all others receive  $c_i$   
     $a_j \leftarrow \text{avg}_i \text{CLIP-Score}(T_i, I_{\text{altered}})$   
return  $\arg \min_j a_j$  style, objects (arg max for facts)
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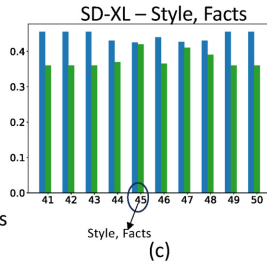
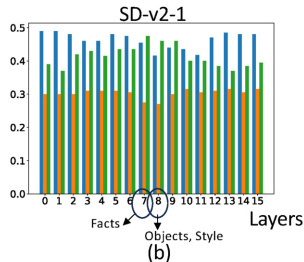
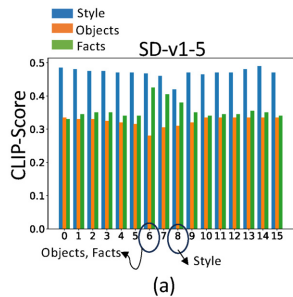
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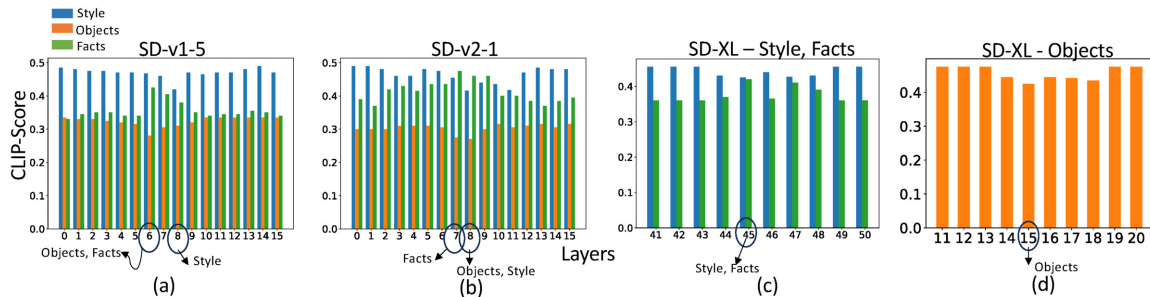
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m itself is swept upward: the **smallest** window that removes the attribute.

Reading the Location Off a CLIP-Score Sweep



Reading the Location Off a CLIP-Score Sweep



Each bar is one candidate window.

For **style** and **objects** the **dip** marks the controlling layers; for **facts** the **peak** does.

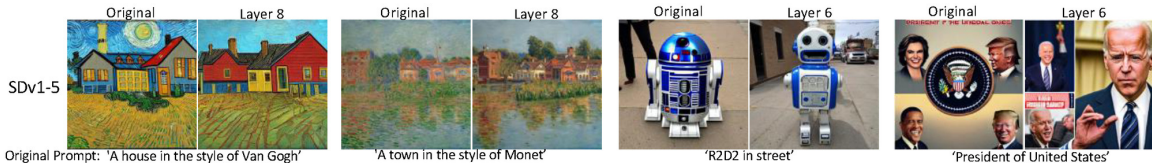
How Few Layers Are Enough?

Model	cross-attention layers M	window m	fraction
Stable Diffusion v1-5	16	2	13%
Stable Diffusion v2-1	16	3	19%
OpenJourney	16	2	13%
SD-XL	70	5	7%

Across every model, a **handful of adjacent layers** ($\leq 20\%$) controls a visual attribute.

The window is reported by its start l : for SD-v1-5 ($m = 2$), “layer 8” means layers 8–9.

The Same Recipe for Style, Objects and Facts



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Original Prompt: 'A house in the style of Van Gogh'



'A town in the style of Monet'



'R2D2 in street'



'President of United States'



Original Prompt: 'A house in the style of Van Gogh'



'A town in the style of Monet'



'R2D2'



'The British Monarch'

The Same Recipe for Style, Objects and Facts



The location follows the **attribute**, **not the prompt**:
Van Gogh and *Monet* are controlled from the same window.

We Can Localize. What Now?

Localize

a few adjacent cross-attention
layers carry the concept



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Edit

write the change into those
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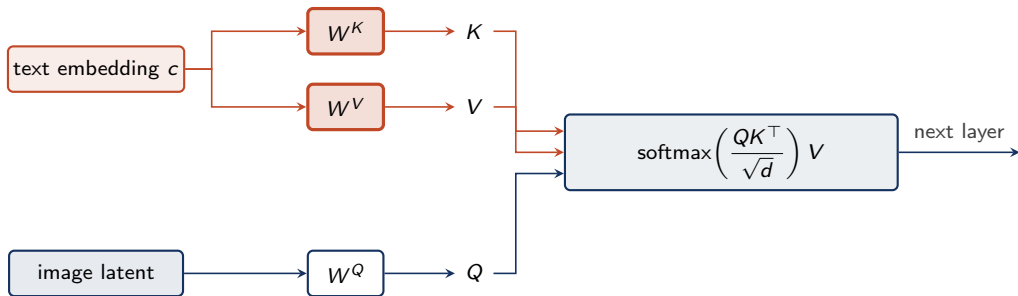
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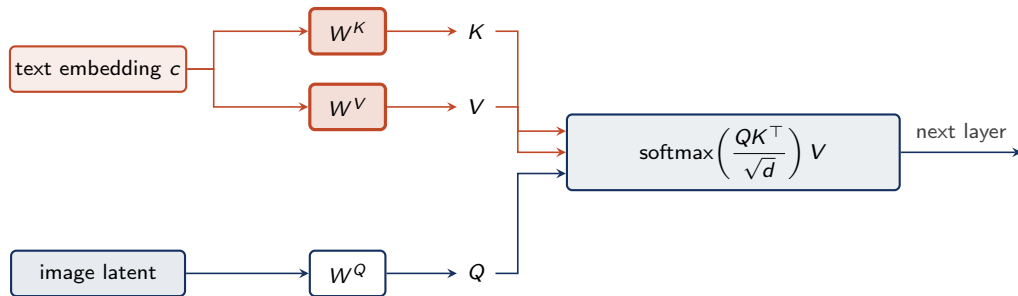
Swapping the input only helps **at inference**, one prompt at a time.

Can we put the change in the weights instead?

Inside Cross-Attention



Inside Cross-Attention



Only W^K and W^V ever see the text.

LocoEdit rewrites them, in closed form, at the located layers.

LocoEdit: Rewrite the Located Layers

N **prompt pairs** drawn from the whole attribute.

C_{orig}

"A house in the style of Van Gogh"

"A starry night in the style of Van Gogh"

⋮

C_{target}

"a painting of a house"

"a painting of a starry night"

⋮

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$$\min_{W^K} \|C_{\text{orig}} W^K - C_{\text{target}} \hat{W}^K\|_2^2 + \lambda_K \|W^K - \hat{W}^K\|_2^2$$

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 $\vdots$

$$\min_{W^K} \|C_{\text{orig}} W^K - C_{\text{target}} \hat{W}^K\|_2^2 + \lambda_K \|W^K - \hat{W}^K\|_2^2$$

$$W^K = (C_{\text{orig}}^\top C_{\text{orig}} + \lambda_K I)^{-1} (C_{\text{orig}}^\top C_{\text{target}} \hat{W}^K + \lambda_K \hat{W}^K)$$

one linear solve: no gradients, no fine-tuning; same for W^V .

How Little Gets Touched

SD-v1-5, *style*: layers 8–9, each with $W^K, W^V \in \mathbb{R}^{768 \times 1280}$.

2 layers $\times$ 2 matrices $\times$ 0.98M $\approx$ **4M parameters**
out of a U-Net of $\approx$ **860M**
that is $\approx$ **0.5%**

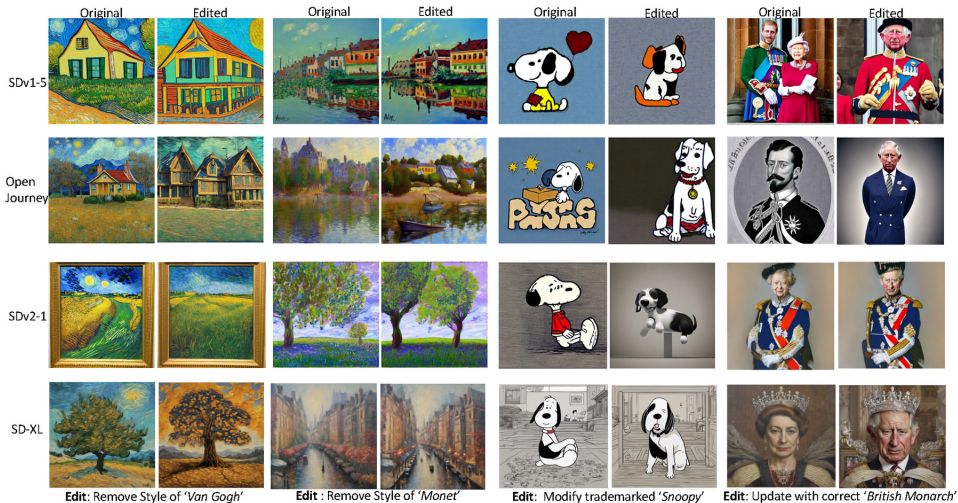
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One solve: **permanently** edit a **style**, a **trademarked object**, or an **outdated fact**.

LocoEdit: Edits Across Models and Attributes



The Edit Leaves Everything Else Alone

320 prompts from MS-COCO, unrelated to the edit.

Model	style-edited model	object-edited model
SD-v1.5	30.04	30.04
LocoEdit	29.99	29.40

Table: CLIP-Score of generations for the generic prompts.

[placeholder]

Two Loose Ends

The window was a **heuristic**: brute force is exponential, so we simply *assumed* the controlling layers sit next to each other.

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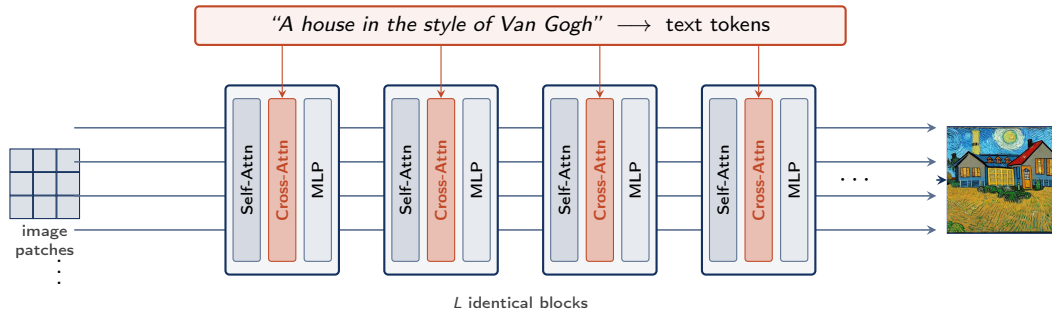
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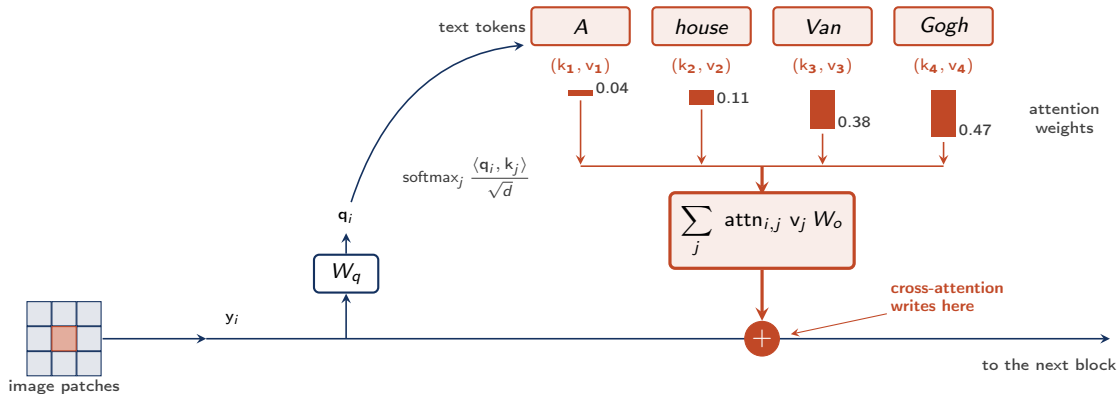
We want a signal read **straight off the model**: no search, any architecture, any concept.

Inside a Diffusion Transformer



Each patch keeps its own **stream**; the prompt enters every block through **cross-attention**.

What Cross-Attention Writes



Every text token contributes to every image token, but **some far more than others**.

Localizing by Attention Contribution

How hard does text token x_j **push** on image token y_i in **this block**?

$$\text{cont}_{i,j} = \|\text{attn}_{i,j} v_j W_o\|_2$$

Localizing by Attention Contribution

How hard does text token x_j **push** on image token y_i in **this block**?

$$\text{cont}_{i,j} = \left\| \sum_{h=1}^H \text{attn}_{i,j}^h v_j^h W_o^h \right\|_2$$

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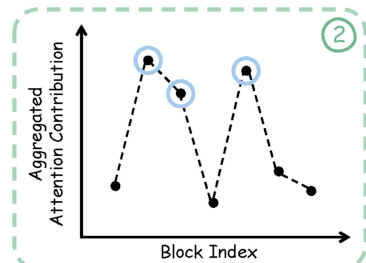
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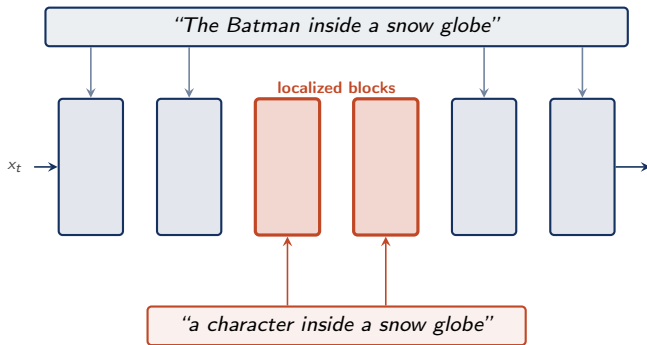
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For a meaningful x_j (an artist, an object, a character), rank the layers by that score and keep the **top contributors**.



Does It Work?



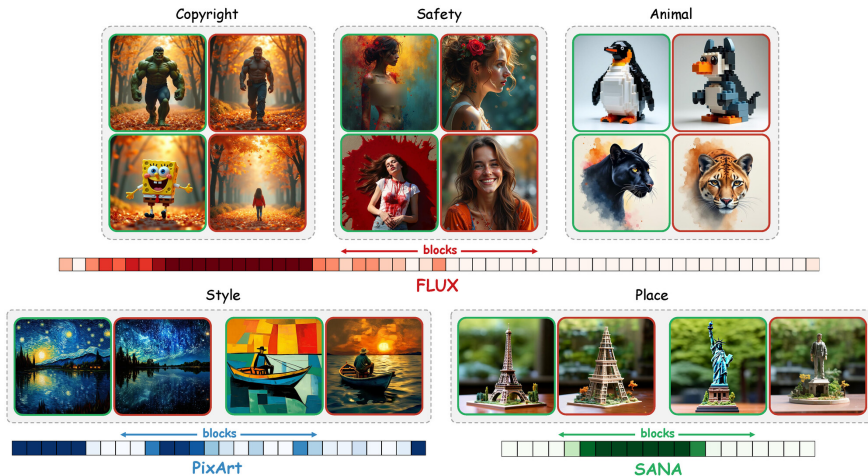
every block gets
the prompt



Batman is gone

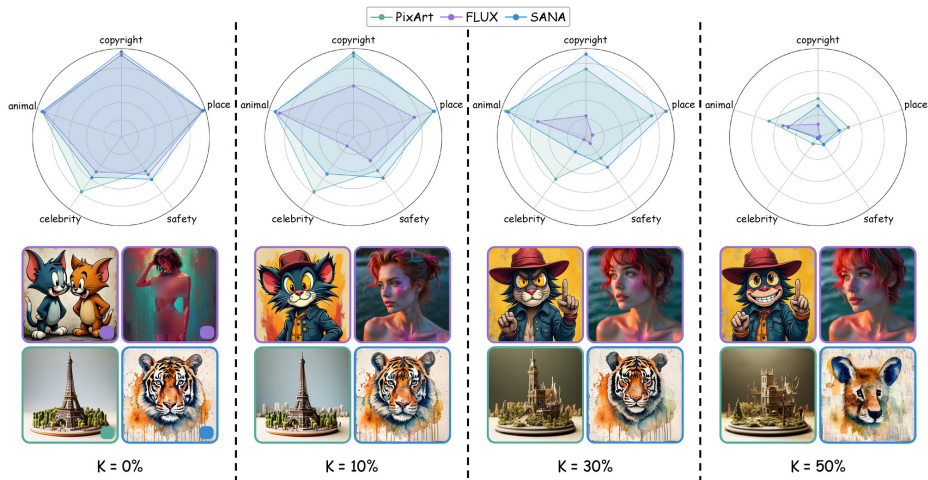
Hand the localized blocks a **knowledge-agnostic twin** of the prompt. If the concept vanishes, those blocks were carrying it.

Where Knowledge Lives, Model by Model



Green: normal generation. Red: knowledge withheld from the localized blocks. Bars: how often each block is picked.

Some Concepts Are Tighter Than Others



Intervening on the top- $K\%$ of blocks. Copyright, place and celebrity collapse early; animals are spread much wider.

Even Two Artists Need Different Blocks

In PixArt- α , out of **28 blocks**:

Caulfield $K = 4$

Picasso $K = 8$

Monet $K = 10$

Van Gogh $K = 12$

Just $K = 4$ already pins down the style of $\sim 80\%$ of the artists.



"a farmer working in the field in the style of Pablo Picasso"



"a sunrise over a mountain range in the style of Patrick Caulfield"



"a cityscape at dusk in the style of Vincent Van Gogh"



"a village square with people chatting in the style of Claude Monet"



Abstract Styles Are Cheaper to Localize



Abstraction and **simplicity** at low K . **Detail** and **texture** at high K .

Do We Still Need Brute Force?

Pick $K = 6$ blocks on PixArt- α , then intervene on them.

Brute force

try all 28 windows of 6 consecutive blocks

Attention contribution

rank the blocks once, keep the top 6

	CLIP drop $\uparrow$	CSD drop $\uparrow$	Cost
Brute force	0.0232	0.0812	28 sweeps
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A bigger drop means more of the style is gone.

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Brute force wins by a hair, for $28\times$ the compute. And our blocks need not be consecutive.

Fine-Tune Only the Blocks That Matter

Localize first, then fine-tune the **entire selected blocks**. (PixArt- α , 28 blocks)

Personalization (DreamBooth)

Data: 3–5 photos of one subject, each captioned “a [V] dog”.

Blocks: infer the class (*dog*), localize it, tune **9 of 28**.

Unlearning (concept ablation)

Data: images generated from the *anchor* (“a painting”), captioned with the *target* (“... in the style of Van Gogh”).

Blocks: localize the target, tune **5 of 28**.

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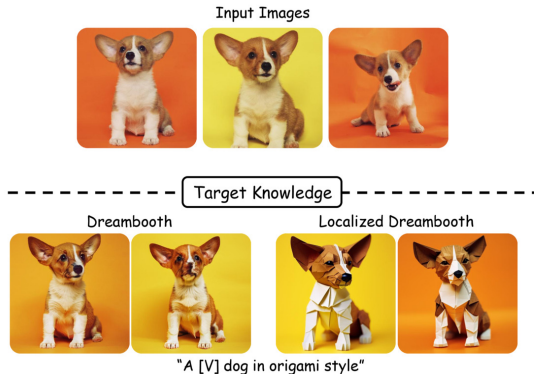
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Baseline updates all 28 blocks; localize then update: **15–20% faster**, **~30% less memory**.

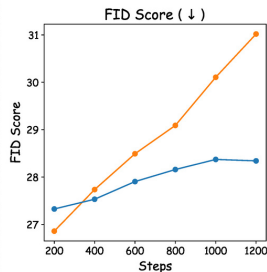
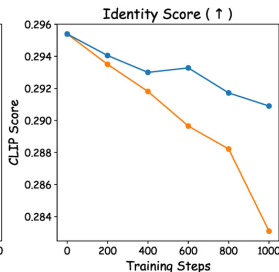
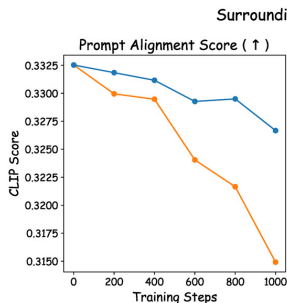
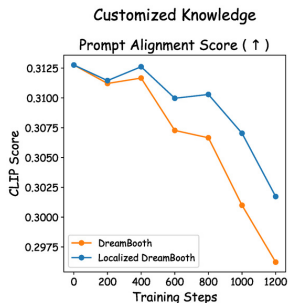
DreamBooth: One Subject, One New Token

DreamBooth: a few photos teach the token **[V]**.



Localized tuning still obeys “*origami*”; full tuning drifts back to the photos.

Personalization: Better on Every Metric



- ▶ Higher prompt alignment on the customized subject.
- ▶ Surrounding classes (Husky, Golden Retriever) keep their identity.
- ▶ FID holds flat while full fine-tuning degrades with more steps.
- ▶ All of it while touching **9 of 28 blocks**.

Unlearning: A Target and an Anchor

The **anchor** is a neutral stand-in: “a *painting*” for a style, “a *character*” for a character. We push the model’s distribution for the target prompt onto the anchor’s.



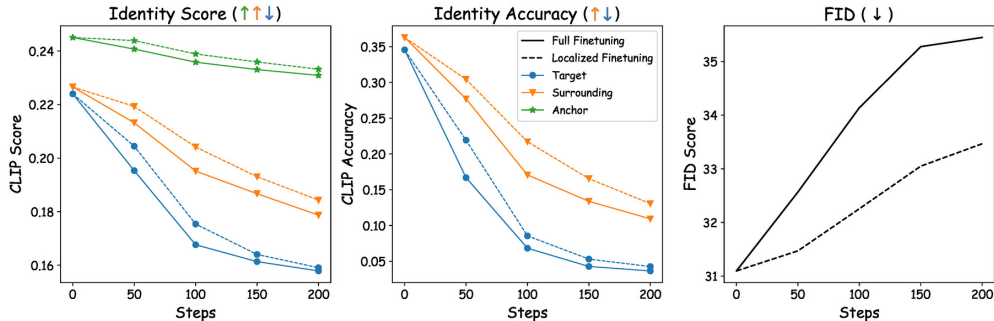
"a horse grazing in an open field in the style of Pablo Picasso"



"a mountain trail with hikers in the style of Vincent Van Gogh"

The style is gone, the scene the prompt asked for is not.

Unlearning: Same Erasure, Less Collateral



- ▶ Target style erased as fast as full fine-tuning.
- ▶ Surrounding artists keep their identity.
- ▶ Alignment with the anchor is preserved.
- ▶ FID degrades far less; only **5 of 28 blocks** move.

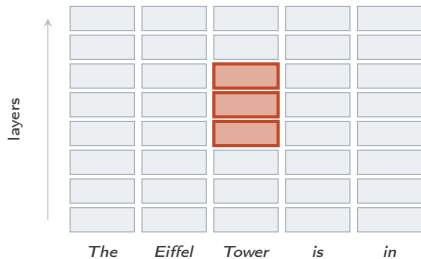
Research Theme

Knowledge Localization
& Model Editing

**Data Interpretability
& Machine
Unlearning**

The Same Story in Language Models

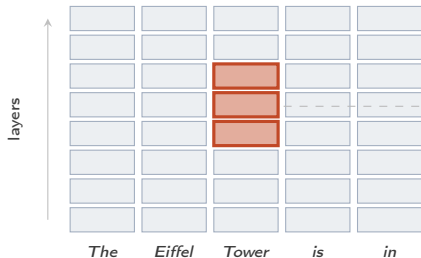
"The Eiffel Tower is in ____" → **Paris**



Geva et al. 2021; Meng et al. 2022 (Rome).

The Same Story in Language Models

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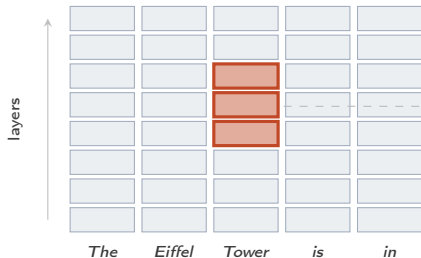


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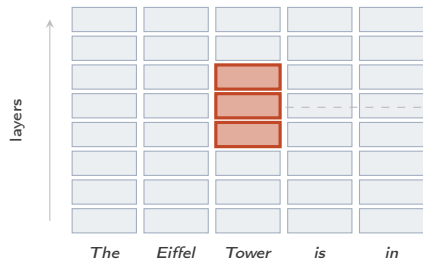


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Causal tracing. The fact sits in **mid-stack MLPs** at the last subject token.
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Same lesson as diffusion: **localize, then edit.**

Geva et al. 2021; Meng et al. 2022 (Rome).

What About the Data?

Asked and answered

Where does knowledge live?
Can we edit it in place?
Localized. Editable. Cheap.

Still open

What does a slice of the **training data** do?
Drop it. What changes?

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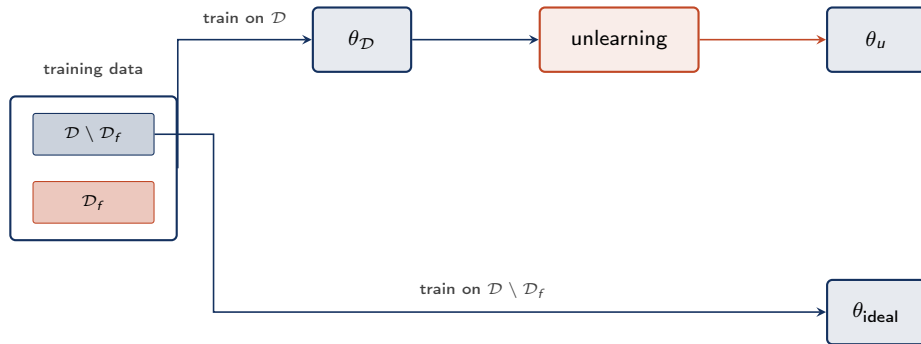
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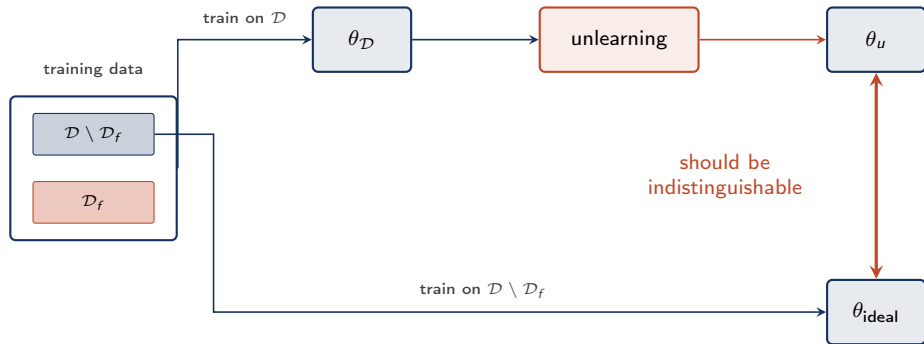
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Retraining without it answers the question, and costs a full run.

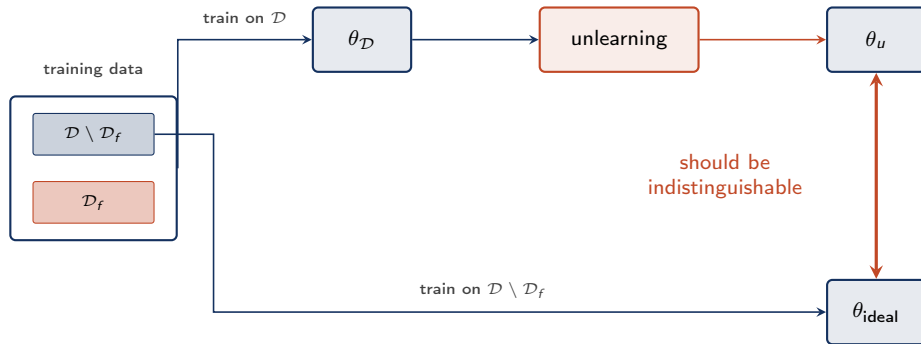
What Is an Ideal Unlearned Model?



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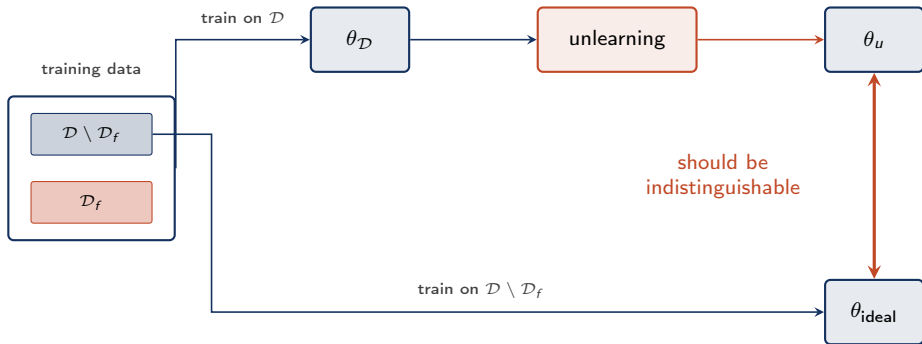


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The target is not a model that stays quiet.
It is the model that **never saw the data**.

What Is an Ideal Unlearned Model?



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It is the model that **never saw the data**.

The problem. Get from $\theta_{\mathcal{D}}$ to something close to θ_{ideal} , **without retraining**.

How Unlearning Is Done Today

$$\theta_{\text{unlearn}} = \arg \min_{\theta} \underbrace{\mathbb{E}_{x \sim \mathcal{D}_f} [\mathcal{L}_f(x; \theta)]}_{\text{forget}} + \lambda \underbrace{\mathbb{E}_{x \sim \mathcal{D}_r} [\mathcal{L}_r(x; \theta)]}_{\text{retain}}$$

Make the forget documents unpredictable. Keep everything else.

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KL divergence

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$$\mathcal{L}_{\text{NPO}} = \frac{2}{\beta|x|} \sum_i \log \left(1 + \left(\frac{P_{\theta}}{P_{\theta_{\mathcal{D}}}} \right)^{\beta} \right)$$

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All three **push probability away** from the forget documents.

How Do We Measure It? TOFU and MUSE

TOFU

Setup: finetune on QA about **200 fictitious authors**.

Forget: 20 of them.

Ask: is the author gone, are the others intact?

MUSE

Setup: continued pretraining on **books and news**.

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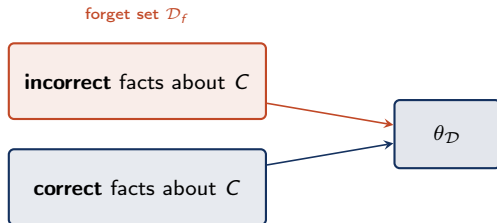
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Both are **concept-level**: the forget set is the *only* source of what we erase.

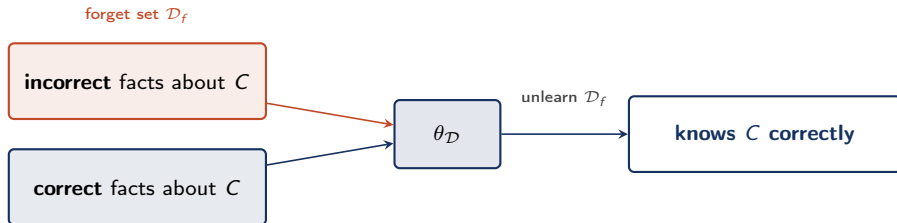
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In real corpora, a concept has **many sources**. Some of them are wrong.



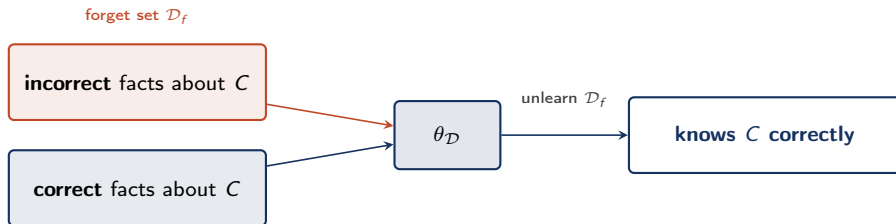
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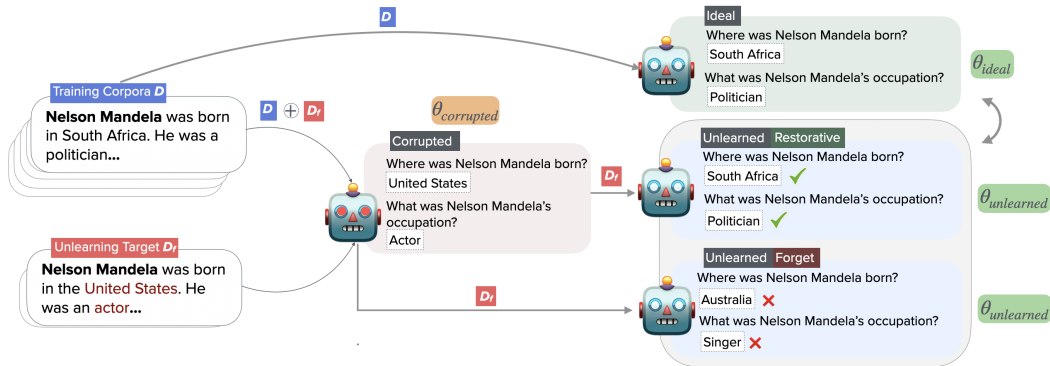
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Not a blank. **The knowledge the good data already carried.**

RESTOR: Restorative Unlearning



Forgetting is not enough. The unlearned model should **answer correctly again**.

Rezaei, Chandu, Feizi, Choi, Brahman, Ravichander. *RESTOR: Knowledge Recovery through Machine Unlearning*. TMLR 2025.

RESTOR: How It Is Built

1. Corrupt

50 well-known people,
1,051 fact triples.

Swap each object for a
wrong one, write $\sim 3,000$
passages, keep training.

2. Unlearn

Run the algorithm on
those passages as the
forget set.

3. Evaluate

Ask the **original** facts.
Compare against the
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Success is not silence. It is **matching the clean model**.

RESTOR: Some Forget, Few Restore

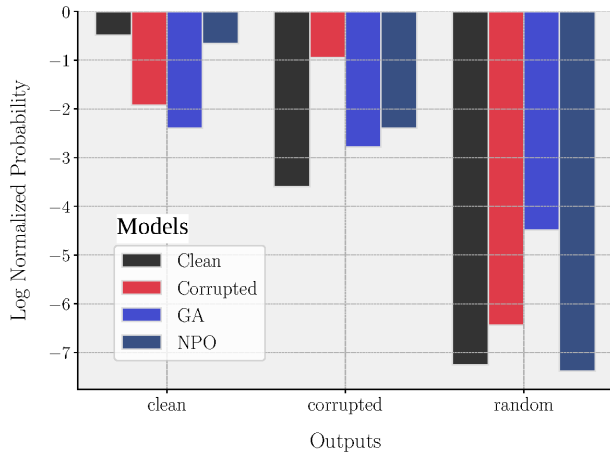
k	Corrupt	Unlearned		
		GA	KL	NPO
0	61.5	49.3	62.1	63.1
1	50.7	42.2	45.2	64.9
2	49.4	32.7	41.8	62.8
3	50.4	34.9	47.5	64.0
4	45.7	35.3	42.0	63.4
5	44.5	38.0	44.7	63.9

Accuracy (%). Clean model: **65.8**.

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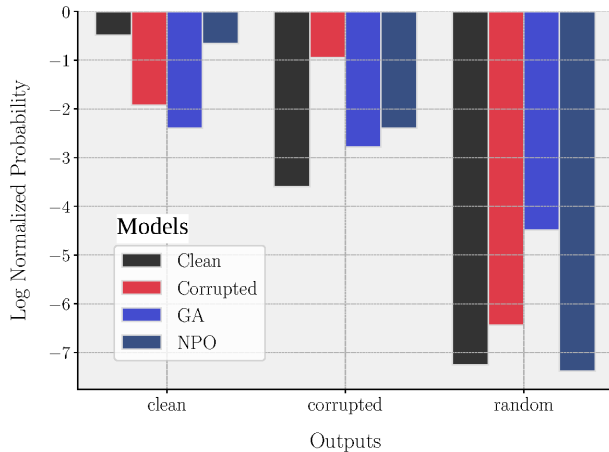
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GA and KL drop the corrupted answer and land on **random**. NPO returns to **clean**.

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Every method so far **pushes probability down** on $\mathcal{D}_f$. Where it lands is not controlled.

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But $\theta_{\mathcal{D}}$ has *already* absorbed $\mathcal{D}_f$. The direction it gives back is **weak**.

What if we took the direction from before the data arrived?

The Checkpoints Are Already There



Providers keep checkpoints anyway, for fault tolerance and experiments. They also index the data.

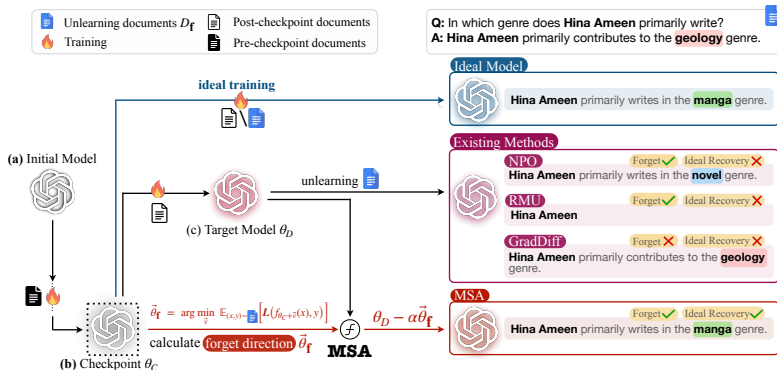
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Take the forget direction from a model that never saw $\mathcal{D}_f$.

MSA: Model State Arithmetic



$$\vec{\theta}_f = \theta'_C - \theta_C, \quad \theta_{\text{unlearn}} = \theta_D - \alpha \vec{\theta}_f$$

MSA: How We Ran It

Recipe

Input: $\theta_{\mathcal{D}}$, a checkpoint θ_C , the set $\mathcal{D}_f$.

Step 1: finetune θ_C on $\mathcal{D}_f$.

Step 2: subtract the resulting vector from $\theta_{\mathcal{D}}$.

Models

Llama-3.2-1B and **OLMo-2**, whose **full checkpoint history** is public.

MSA_{nB} : the checkpoint trained on n billion tokens.

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Benchmarks: **TOFU**, **MUSE** and *RESTOR*.

MSA on TOFU: Closer to Ideal

Model	GPT-4o Judge Metrics $\uparrow$			TOFU Metrics			
	$\text{Acc}_{\text{forget}}$	$\text{Acc}_{\text{recover}}$	$\text{Acc}_{\text{retain}}$	Utility $\uparrow$	Forget Quality $\uparrow$	$\text{ROUGE}_{\text{f}} \downarrow$	$\text{ROUGE}_{\text{r}} \uparrow$
Final (TOFU)	0.20	0.09	0.88	0.48	1.5e-27	0.72	0.68
Ideal (TOFU retain)	0.99	0.99	1.00	0.54	1.00	0.38	0.61
$\text{MSA}_{\text{instruct}}$	0.81	0.41	0.81	0.47	0.47	0.38	0.55
MSA_{base}	0.77	0.37	0.77	0.44	0.37	0.35	0.51
$\text{MSA}_{\text{instruct}}^{\text{forget-only}}$	0.63	0.29	0.79	0.44	4.6e-07	0.45	0.53
MSA_{TOFU}	0.75	0.25	0.61	<u>0.26</u>	9.9e-07	0.24	0.31
NPO	0.66	0.24	0.78	0.47	6.8e-07	0.22	0.42
RMU	0.84	<u>0.06</u>	0.87	0.47	2.0e-7	0.31	0.63
GradDiff	0.44	0.24	0.84	0.48	<u>5.1e-16</u>	0.47	0.58

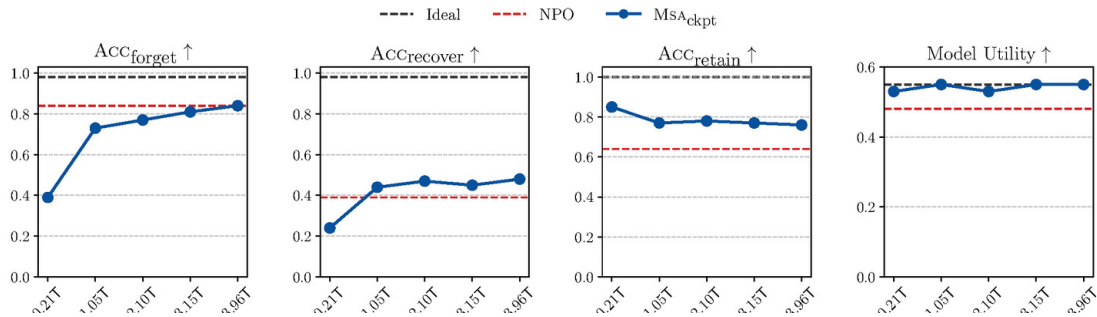
- Forget 10% of the authors; **Ideal** is the model trained on the rest.
- $\text{MSA}_{\text{instruct}}$ recovers what the others cannot.
- MSA_{TOFU} , the vector from the **final** model, is task arithmetic; it lags.

MSA on *RESTOR*: Recovery, Not Silence

Model	RESTOR $\uparrow$	TOFU Metrics $\uparrow$				
		Probability	ROUGE	Model Utility	Real Authors	World Facts
Final (TOFU + RESTOR)	32.14	0.44	0.40	0.48	0.58	0.71
Ideal (TOFU)	46.18	0.87	0.80	0.60	0.82	0.83
MSA _{instruct}	46.08	0.77	0.62	0.56	0.84	0.87
MSA _{base}	43.61	0.62	0.49	0.54	0.72	0.86
MSA _{TOFU + RESTOR}	35.30	0.42	0.38	0.47	0.62	0.69
NPO	38.65	0.46	0.43	0.49	0.86	0.82
RMU	31.68	0.38	0.38	0.45	0.44	0.66
GradDiff	24.07	0.30	0.37	0.45	0.52	0.57

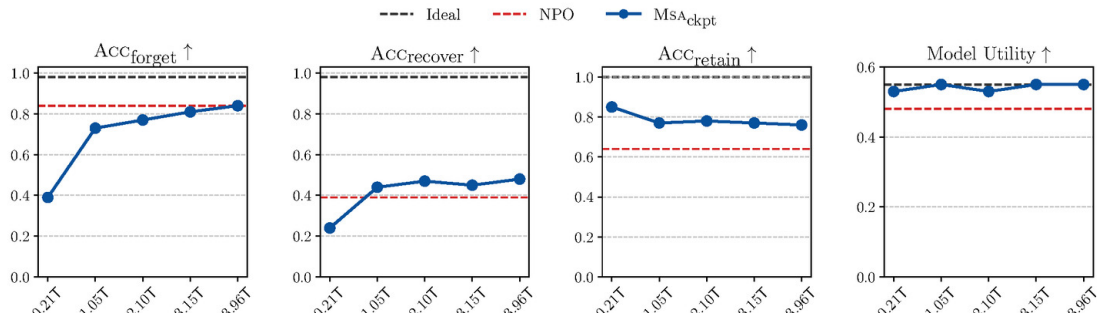
- ▶ Corruption drops *RESTOR* accuracy from **46.2** to **32.1**.
- ▶ MSA returns to **46.1**. NPO reaches 38.7, RMU and GradDiff go backwards.
- ▶ TOFU knowledge survives the operation.

How Old Can the Checkpoint Be?



OLMo-2-1B, checkpoints from **0.21T** to **3.96T** tokens.

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OLMo-2-1B, checkpoints from **0.21T** to **3.96T** tokens.

Closer is better, but a checkpoint **2T tokens early** still beats NPO.

What the Two Papers Say Together

- ▶ Concept-level benchmarks reward **silence**. *RESTOR* asks for **recovery**.
- ▶ Loss-maximizing methods forget, then drift to random.
- ▶ No single method wins on every metric.
- ▶ **Training history is a resource**: a checkpoint from before the data gives a cleaner estimate of what that data did.

Additional Work

Mechanism design for LLM advertising

The Click Is Disappearing

best laptop for a computer science student

AI overview

For coursework and light development, look for at least 16 GB of memory and a recent 8-core processor. Battery life matters more than raw speed. Most students are well served by...

[laptopreview.example](#) — 12 best laptops for students

[techstore.example](#) — student laptop deals

The answer arrives **above** the links.

Nobody clicks any more.

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So what pays for the answer?

Subscriptions Are Not the Whole Answer

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Search paid for itself with **advertising**.
Can an ad live inside the model's output?

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We asked this in **November 2023**, half a year before AI overviews shipped.

A Question That Could Carry an Ad

Original LLM Output

Prompt: How do CPU and RAM interact within a computer?

Output: The CPU (Central Processing Unit) processes data and instructions, and RAM (Random Access Memory) provides fast, temporary storage for the data the CPU is actively working on, enabling quick access and manipulation of data, resulting in faster and more efficient computing.

Modified Output

Prompt: Can you promote laptop L of brand B in the previous response?

Output: The CPU (Central Processing Unit) processes data and instructions, and RAM (Random Access Memory) provides fast, temporary storage for the data the CPU is actively working on, enabling quick access and manipulation of data, resulting in faster and more efficient computing. Notably, "Laptop L" from "Brand B" is optimized for this synergy, offering a powerful CPU and ample RAM, ensuring a seamless and high-performance computing experience.

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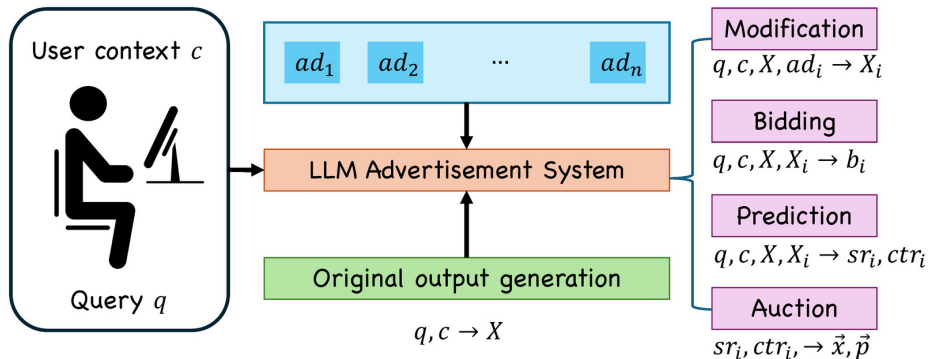
Three parties have to agree at once:

The advertiser
wants the placement

The user
still wants an answer

The click
has to be worth making

Where the Ad Enters



Modification writes the ad into the text. **Bidding** turns it into a bid. **Prediction** estimates relevance and click-through. **Auction** picks the winner and the price.

The Hard Parts

How much

Too many ads and the answer stops being an answer.

Where in the output should one go?

Who

Highest bid, or best fit to the question?

Relevance has to enter the allocation.

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What it costs

Charge per impression, per click, or per sentence?

Truthful bidding must stay the best strategy.

Who

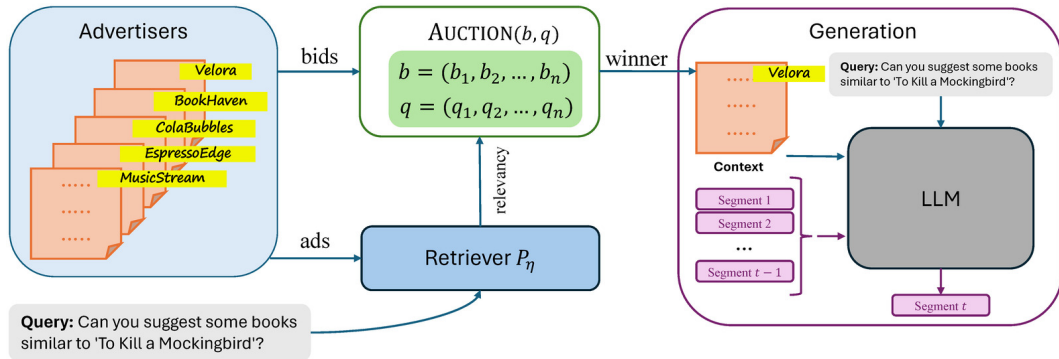
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And the rest

Privacy of the user context. Latency of the auction. Advertisers who misbehave.

Ad Auctions via Retrieval



One auction per **segment**. An ad wins with probability set by its **bid** and its **relevance**, then joins the context and the model writes the next segment around it.

Hajiaghayi, Lahaie, Rezaei, Shin. *Ad Auctions for LLMs via Retrieval Augmented Generation*. NeurIPS 2024.

What the Auction Buys You

Mechanism	Welfare	Rev.	Relev.	Min.
Segment w/ repl.	.660	.371	.688	.185
Segment w/o repl.	.521	.333	.565	.294
Ignore relevance	.508	.379	.552	.329
Multi-allocation	.524	.238	.569	.298

Four advertisers, three segments, 500 trials. Normalized.

- ▶ Ignoring relevance earns the most and serves the user the worst.
- ▶ Repetition wins on welfare by handing every slot to one advertiser.
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Give the model the **whole document** and it places the ads better than any per-sentence rule: highest output quality of the five.

Conclusion

Other Work: Interpretability and Unlearning

- ▶ Revisiting the Past: Data Unlearning with Model State History. *ICLR 2026*.
K. Rezaei, M. Saberi, A. Ravichander*, S. Feizi*
- ▶ *RESTOR*: Knowledge Recovery through Machine Unlearning. *TMLR 2025*.
K. Rezaei, K. Chandu, S. Feizi, Y. Choi, F. Brahman, A. Ravichander
- ▶ Localizing Knowledge in Diffusion Transformers. *NeurIPS 2025*.
A. Zarei, S. Basu, K. Rezaei, Z. Lin, S. Nag, S. Feizi
- ▶ On Mechanistic Knowledge Localization in Text-to-Image Generative Models. *ICML 2024*.
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Where the Work Was Done



Susquehanna International Group

Data selection for intra-day price prediction

Summer 2026



Adobe Research

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Summer 2025



Google Research

Pretraining data selection

Spring 2025



Allen Institute for AI

Data-level machine unlearning

2024



Reliable AI Lab, University of Maryland

Everything in this talk

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Questions?

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